2025.12.3

Thm 1 $\forall\ 1\le p\le\infty,\ k\ge 0\in\mathbb{N}$. $W^{k,p}$ is Banach space.

Pf: ① $\|u\|_{W^{k,p}}$ is a norm. $(1\le p<\infty,\ p=\infty)$

$c\in\mathbb{R}$, $\|c\cdot u\|_{W^{k,p}}=|c|\,\|u\|_{W^{k,p}}$; $\|u\|_{W^{k,p}}=0\Rightarrow\left(\int_\Omega |u|^p dx\right)^{\frac{1}{p}}=0\Rightarrow u=0$ a.e.

$$\|u+v\|_{W^{k,p}}=\left(\int_\Omega \sum_{|\alpha|=0}^{k}|\partial^\alpha(u+v)|^p dx\right)^{\frac{1}{p}}=\left(\int_\Omega\left(\sum_{|\alpha|=0}^{k}|\partial^\alpha u+\partial^\alpha v|^p\right)^{p\cdot\frac{1}{p}}dx\right)^{\frac{1}{p}}$$ (l^p-norm)

$$\le\left(\int_\Omega\left[\left(\sum_{|\alpha|=0}^{k}|\partial^\alpha u|^p\right)^{\frac{1}{p}}+\left(\sum_{|\alpha|=0}^{k}|\partial^\alpha v|^p\right)^{\frac{1}{p}}\right]^p dx\right)^{\frac{1}{p}}$$ (L^p-norm)

$$\le\left(\int_\Omega\sum_{|\alpha|=0}^{k}|\partial^\alpha u|^p dx\right)^{\frac{1}{p}}+\left(\int_\Omega\sum_{|\alpha|=0}^{k}|\partial^\alpha v|^p dx\right)^{\frac{1}{p}}$$

$$=\|u\|_{W^{k,p}}+\|v\|_{W^{k,p}}$$

② Completeness of $\|\cdot\|_{W^{k,p}}$

Suppose $\{u_m\}$ is Cauchy sequence in $W^{k,p}(\Omega)$

$\Rightarrow$ $\{u_m\}$ is Cauchy sequence in $L^p(\Omega)$, $\{\partial^\alpha u_m\}$ is Cauchy sequence in $L^p(\Omega)$ $\xrightarrow{\text{completeness of } L^p}$ $u_m\xrightarrow{L^p}u_\infty$, $\partial^\alpha u_m\xrightarrow{L^p}v_\alpha$

Claim: $v_\alpha=$ weak $\partial^\alpha u_\infty$ $\forall|\alpha|\le k$.

$\forall\varphi\in C_c^\infty(\Omega)$,

$$\int_\Omega \partial^\alpha u_m\cdot\varphi dx=(-1)^{|\alpha|}\int_\Omega u_m\cdot\partial^\alpha\varphi\,dx \quad m\to\infty$$

$$\int_\Omega v_\alpha\cdot\varphi dx=(-1)^{|\alpha|}\int_\Omega u_\infty\cdot\partial^\alpha\varphi dx$$

$$\int_\Omega|\partial^\alpha u_m-v_\alpha|\cdot|\varphi|dx\overset{\text{Hölder}}{\le}\left(\int_\Omega|\partial^\alpha u_m-v_\alpha|^p dx\right)^{\frac{1}{p}}\left(\int_\Omega|\varphi|^{p'}dx\right)^{\frac{1}{p'}}\to 0\ (m\to\infty)$$

$\Rightarrow v_\alpha=$ weak $\partial^\alpha u_\infty\Rightarrow u_\infty\in W^{k,p}(\Omega)$ & $\|\|u_m-u_\infty\|\|_{W^{k,p}(\Omega)}\to 0\ (m\to\infty)$

$\Leftrightarrow\|u_m-u_\infty\|_{W^{k,p}(\Omega)}\to 0$

$\Rightarrow W^{k,p}$ is complete.

Thm 2 $W^{k,p}(\Omega)$ is separable if $1\le p<\infty$; $\Leftrightarrow L^p$ is separable $1\le p<\infty$

$W^{k,p}(\Omega)$ is reflexive if $1<p<\infty$ L^p is reflexive $1<p<\infty$

X is a Banach space. $X^*=\{u^*: X\to\mathbb{R}\mid$ linear & bounded$\}\to$ dual space

of X. Banach. $J: X \to (X^*)^* = X^{**}$

$x \mapsto J(x): X^* \to \mathbb{R}$ linear & bounded

$\langle J(x), u^* \rangle = \langle u, x \rangle$

If $J(X) = X^{**}$, then X is reflexive → weak compactness:

$\forall \{x_n\} \subset X$ bounded, $\exists$ subsequence $\{x_{k_j}\}_{j=1}^{\infty} \subset \{x_k\}_{k=1}^{\infty}$ & $x_\infty \in X$ s.t.

$x_{k_j} \xrightarrow{w} x_\infty$ $(j \to \infty)$ (i.e. $u^* \in X^*, \langle u^*, x_{k_j} \rangle \to \langle u^*, x_\infty \rangle$ $j \to \infty$)

Pf: Define: $T: W^{k,p}(\Omega) \to \prod_{|\alpha|=0}^{k} L^p(\Omega) := Y$

$u \mapsto \{\partial^\alpha u\}_{|\alpha|=0}^{k} = \{u, \partial_{x_1} u, \cdots, \partial_{x_n} u, \partial^2_{x_i x_j} u \cdots\}$.

norm: $\|Tu\|_Y = \|u\|_{W^{k,p}(\Omega)} \Rightarrow T$ → linear & isometry

$\Rightarrow W^{k,p}(\Omega)$ has the same properties as $T(W^{k,p}(\Omega))$, which is a closed linear subspace of Y, and hence has the same properties as Y.

Def 1 (Partition of unity)

Let $\{U_n\}_{n=1}^{\infty}$ be bounded & open sets in Ω s.t. (*1)

(1) $\bar{U}_n \subset \Omega$; (2) $\forall$ compact subset K in Ω intersects only finitely U_n's

(3) $\bigcup_{n=1}^{\infty} U_n = \Omega$

A partition of unity subordinate to the covering $\{U_n\}$ of Ω is $\{\phi_n\}_{n=1}^{\infty} \subset C_c^\infty(\Omega)$ s.t.

(a) $\phi_n \geq 0$; (b) $\operatorname{supp} \phi_n \subset U_n$; (c) $\sum_{n=1}^{\infty} \phi_n(x) = 1$. $\forall x \in \Omega$

Thm3

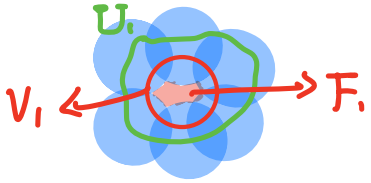
Given a covering $\{U_n\}$ of Ω that satisfies (*1),

a partition of unity subordinate to it always exists.

pf: Step1. Construct a new open covering $\{V_n\}$ s.t. $\overline{V_n} \subset U_n$, $\bigcup_{n=1}^{\infty} V_n = \Omega$

Let $F_1 = \overline{U_1} \setminus \bigcup_{n=2}^{\infty} U_n$ closed

$F_1 \subset U_1 \Rightarrow dist(F_1, \partial U_1) > 0$



Define $V_1 = \{x \in U_1 : dist(x, \partial F_1) < \frac{1}{2} dist(F_1, \partial U_1)\}$ → open, $F_1 \subset V_1$

$\overline{V_1} \subset U_1$, $V_1 \cup (\bigcup_{n=2}^{\infty} U_n) = \Omega$

Let $F_2 = \overline{U_2} \setminus (V_1 \cup \bigcup_{n=3}^{\infty} U_n) \Rightarrow F_2$ closed, $F_2 \cup (\bigcup_{n=3}^{\infty} U_n \cup V_1) = \Omega$, $F_2 \subset U_2$

$\Rightarrow \exists V_2$ open s.t. $F_2 \subset V_2 \subset \overline{V_2} \subset U_2 \Rightarrow V_1 \cup V_2 \cup (\bigcup_{n=3}^{\infty} U_n) = \Omega$. ....

Step2. Construct $\psi_n \in C_c^{\infty}(U_n)$ s.t. $\psi_n \geq 0$, $\psi_n > 0$ on $\overline{V_n}$, $\forall n \geq 1$

Let $\psi_n = (\mathbb{1}_{V_n})_\varepsilon = \int_\Omega j_\varepsilon(x-y) \mathbb{1}_{V_n}(y) dy = \int_{V_n} j_\varepsilon(x-y) dy$, $\mathbb{1}_{V_n} \in L^1(\Omega)$

$\Rightarrow \psi_n \in C_c^{\infty}(\mathbb{R}^n)$, $supp\, \psi_n(x) \subset U_n$ if $\varepsilon > 0$ small.

$\psi_n(x) > 0$ $x \in \overline{V_n}$ ($Leb(V_n \cap B_\varepsilon(x)) > 0$)

Step3. $\psi(x) = \sum_{n=1}^{\infty} \psi_n(x)$. $\forall$ fixed $x_0 \in \Omega = \bigcup_{n=1}^{\infty} V_n \Rightarrow \exists n$. $x_0 \in V_n$

$\Rightarrow \psi(x_0) \geq \psi_n(x_0) > 0$

Take small $\delta > 0$, s.t. $\overline{B_\delta(x_0)} \subset \Omega \Rightarrow$ only finitely many U_n intersect $\overline{B_\delta(x_0)}$

$supp\, \psi_n \subset U_n \Rightarrow$ On $\overline{B_\delta(x_0)}$, $\sum_{n=1}^{\infty} \psi_n(x)$ is a finite sum.

$\Rightarrow$ on $\overline{B_\delta(x_0)}$, $\psi(x)$ is well-defined & $\psi \in C^{\infty}(\overline{B_\delta(x_0)})$

$\Rightarrow \psi \in C^{\infty}(\Omega)$, $1 = \frac{\psi(x)}{\psi(x)} = \sum_{n=1}^{\infty} \frac{\psi_n(x)}{\psi(x)} := \sum_{n=1}^{\infty} \phi_n(x)$